

CHAPTER 23

TPE331-10 ENGINE LIFT FIXTURE (J-HOOK)

23.1 GENERAL INFORMATION.

The TPE331-10 engine lift fixture is used to lift the engine for removal and installation on the MQ-9 aircraft. The engine lift fixture can also be used to place the engine in a shipping and storage container. See Figure 38-94 for illustrated parts breakdown.

23.2 PURPOSE.

The engine lift fixture is an L-shape lifting fixture that attaches to the engine with an anti-rotation feature. There are several attachment points along the spine of the lift hook along with two stabilizing lanyard rings.

23.3 SUPPORT EQUIPMENT.

The following is a list of support equipment used to maintain the engine lift fixture:

Table 23-1. Support Equipment.

Name	Identification	Quantity
Torque wrench, 40-200 in-lbs	Local purchase	1

23.4 CONSUMABLES.

There are no specific consumables required for maintaining the engine lift fixture.

23.5 PRE-USE INSPECTION.

WARNING

Do not exceed maximum load limit of 750 pounds. Failure to comply may result in injury to personnel and/or damage to aircraft and equipment.

Refer to Figure 23-1 for lift fixture pre-use inspection.

- Inspect overall condition for serviceability.
- Verify identification label is properly attached to lift fixture.
- Inspect hardware for security.

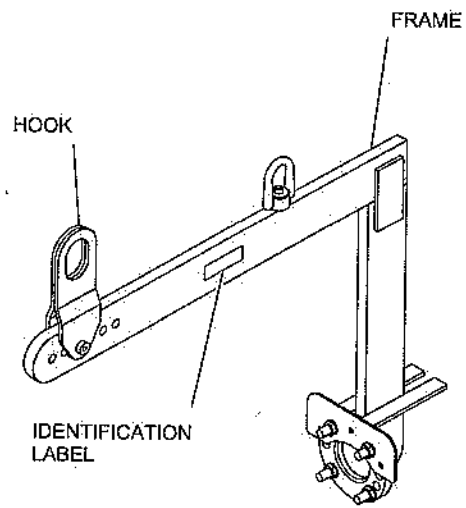


Figure 23-1. TPE331-10 Engine Lift Fixture (J-Hook).

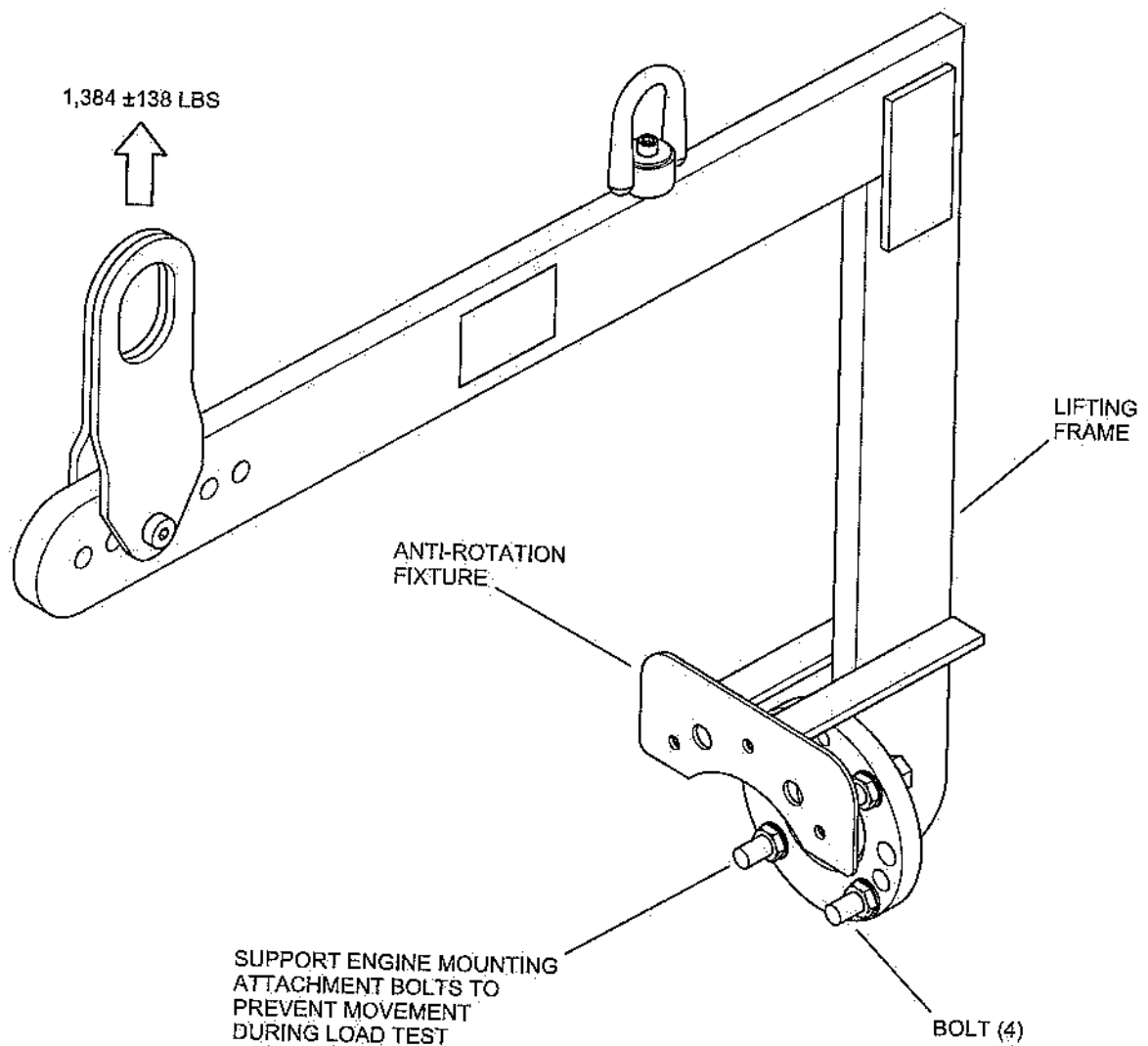


Figure 23-2. TPE331-10 Engine Lift Fixture (J-Hook) Load Test.

23.6 ANNUAL INSPECTION.

Refer to Figure 23-2 for TPE331-10 Engine Lift Fixture (J-Hook) load test.

WARNING

New, extensively repaired, and/or altered equipment shall be load tested by authorized personnel in accordance with applicable manufacturer's documentation and AFI 91-203 before use. Use care while performing load test. Failure to comply may result in injury to personnel and/or damage to equipment.

- Refer to DISASSEMBLY and remove anti-rotation fixture from J-Hook lifting frame.
- Support lift fixture engine mounting attachment bolts to prevent movement during load test.
- Apply 1,384 ± 138 pounds to lifting clevis in direction shown.

- d. Hold load for 30 seconds.
- e. Remove load and inspect for damage.
- f. Perform Nondestructive Inspection (NDI) on welds in accordance with TO 33B-1-2. If cracks are found, repair in accordance with TO 00-25-252.

23.7 DISASSEMBLY.

Refer to Figure 23-3 for disassembly of engine lift fixture.

- a. Remove hoisting ring from lifting frame.
- b. Remove one bolt, one washer, one nut, and lifting clevis from lifting frame.
- c. Remove two bolts, four washers, two nuts, and anti-rotation fixture from lifting frame.

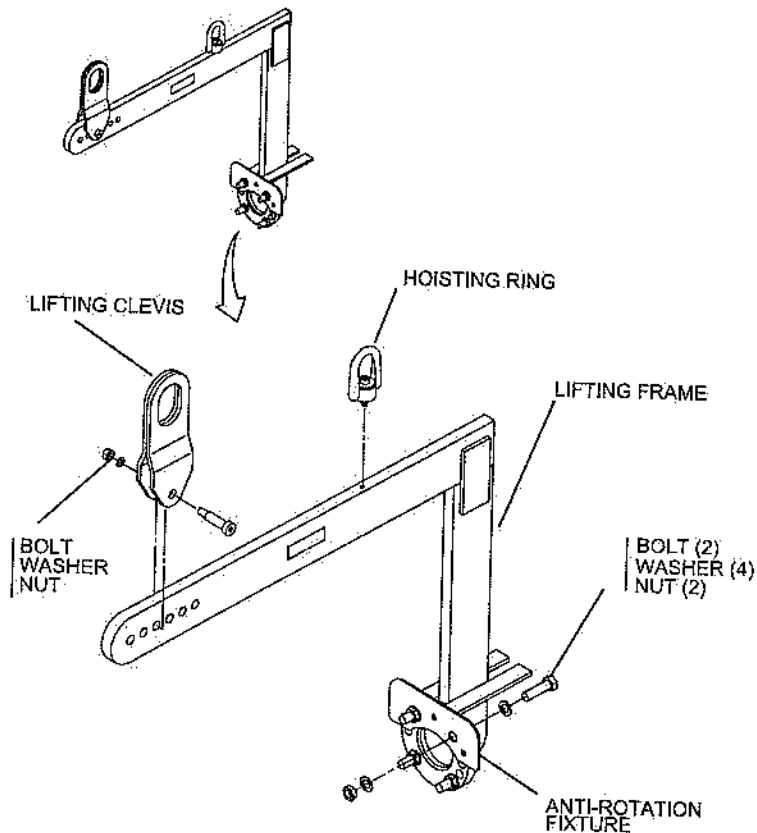


Figure 23-3. TPE331-10 Engine Lift Fixture Assembly.

23.8 ASSEMBLY.

Refer to Figure 23-3 for assembly of lift fixture.

- a. Install two bolts, four washers, and two nuts to secure anti-rotation fixture to lifting frame.
- b. Install one bolt, one washer, and one nut to secure lifting clevis to engine lifting frame.

- c. Torque nut to 50 ± 2 in-lbs.
- d. Install hoisting ring on lifting frame.
- e. Torque hoisting ring to 60 ± 2 in-lbs.